Monocular Depth Estimation: A Survey

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***Abstract*—Monocular depth estimation is an ill-posed task in computer vision, which holds great significance in the fields such as artificial intelligence, virtual reality, augmented reality, path planning, unmanned driving, and navigation guidance. The primary objective of monocular depth estimation is to predict the depth value of each pixel or infer depth information, given just a single red-green-blue (RGB) image as input. Traditional monocular depth estimation methods rely on limited depth cues, such as strict scene conditions. With the significant advancements in computer vision and artificial intelligence, monocular depth estimation using deep learning has been extensively researched and has yielded substantial results. This paper presents a comprehensive survey of monocular depth estimation. Firstly, we give an overall introduction to monocular depth estimation and explain it from traditional and deep learning-based methods, respectively. To specify, supervised, self-supervised and semi-supervised models are described in detail in deep learning-based methods. Additionally, we introduce publicly available benchmark datasets and evaluation metrics commonly used in this field. Finally, we discuss the current challenges and promising prospects for the development of monocular depth estimation.**

***Index Terms*—Monocular depth estimation, Deep learning, Supervised model, Self-supervised model, Semi-supervised model**

I. Introduction

Monocular depth estimation is the task of inferring the depth value for each pixel in a single or monocular RGB image, which has significant applications in scene understanding and three-dimensional (3D) reconstruction. By combining the depth information with camera motion data, it becomes possible to reconstruct a 3D representation of the environment. The acquisition of dense depth maps can be leveraged for various purposes including but not limited to path planning, robot navigation, autonomous driving, virtual reality, augmented reality, and the development of other visual tasks [1]–[3]. Estimating depth from a single image offers distinct advantages, particularly in scenarios where other sources of information such as stereo images, optical flow, and point clouds are unavailable or insufficient for accurate depth recovery [4].

The acquisition of depth information can be categorized into active and passive modes. The active depth estimation methods typically rely on lasers, structured light, and other reflections on the object surface to acquire depth information and estimate scene depth maps [5]. Currently, the most commonly used devices comprise laser radar, time-of-flight (TOF) cameras, and Kinect cameras. Laser radar calculates the target distance by measuring the time difference between laser pulse emission and reception. Though this method boasts high measurement accuracy, it yields extremely sparse depth maps. The KITTI dataset [6] is collected by the laser radar, with depth information points being captured alongside the entire image. The number of points does not exceed 5%, and the dense depth map can be obtained through the completion algorithm. The working principle of the TOF camera is similar to that of the laser radar. The synthetic light emitted by the former serves for distance measurement, yet its imaging resolution is relatively low and the edges of object details are prone to loss, making long-distance imaging difficult. The Kinect camera is a 3D RGB depth sensor released by Microsoft Corporation [7]. Despite the high sensing performance and real-time functionality of Kinect cameras, the effective distance is limited to 0.8-4.5 meters, which restricts the application in outdoor fields.

The passive depth acquisition methods are based on software algorithms, using the features of two-dimensional images for depth estimation or depth completion. This method does not require the heavy cost of manpower or computing resources [8]. In an industrial setting, the execution of a specific task necessitates the acquisition of multi-depth information. For instance, in the scenario of a robot grasping an object, it is imperative to obtain depth information for constructing environmental maps as well as determining the distance between the robot gripper and the target object. If all depth information is acquired through depth cameras, the research and development costs will be high. However, depth estimation only requires the indirect acquisition of depth information through images or videos captured by one or more RGB cameras. Therefore, passive methods for depth acquisition have become the mainstream of research, garnering extensive attention and achieving significant progress in recent years.

Depth estimation algorithms can be implemented in a variety of ways, which can be classified according to the developmental stages of depth estimation algorithms, including depth cues-based algorithms, machine learning algorithms, and deep learning algorithms. The detailed evolution of monocular depth estimation is illustrated in Fig. 1. Depth cues-based algorithms generally use the texture geometric information inside the image to estimate the position information of the camera or restore the depth of the scene through 3D reconstruction [9]–[11]. The depth cues-based algorithms exhibit low accuracy and poor practicality as their general characteristics. Furthermore, these methods are only applicable in restricted scenes [12]–[14]. During the period of machine learning,

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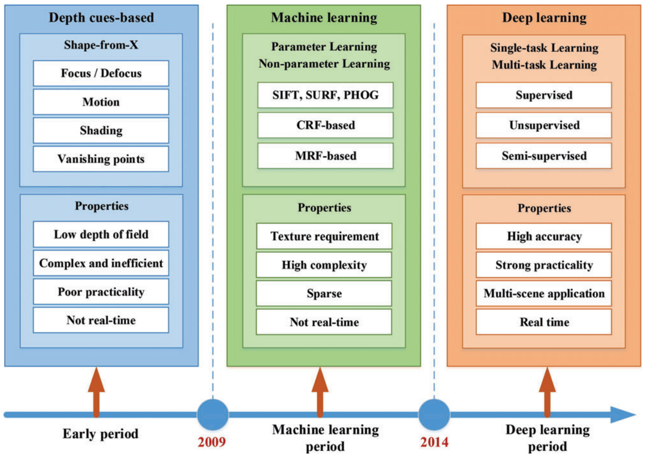


Fig. 1. The detailed evolution of monocular depth estimation, cited from [8]. The evolution can be categorized into three periods: the earlier depth cues-based period, the machine learning period, and the deep learning period.

numerous hand-made features and probabilistic graph models have been proposed for depth estimation modeling, such as pyramid histogram of oriented gradient (PHOG) [15], scale-invariant feature transform (SIFT) [16], speeded up robust features (SURF) [17], conditional random field (CRF) [18] and markov random field (MRF) [19]. With the rapid development of artificial intelligence, many ideas of machine learning algorithms have been extended to deep learning algorithms for depth estimation [20]–[26]. The training of deep neural networks has historically been challenging due to the inevitability of gradient vanishing or explosion upon reaching the deeper layers of the network. ResNet [27] and other achievements have gradually addressed this issue, demonstrating the superiority of deep learning in visual tasks. On the one hand, deep neural networks can acquire structural information for depth inference through supervised signals [28]; on the other hand, the accuracy of deep learning-based methods far exceeds that of traditional algorithms, resulting in superior performance and generalization capabilities.

The survey is structured as follows: Section I provides a general introduction to monocular depth estimation. Section II briefly introduces traditional methods for monocular depth estimation. Section III reviews deep learning-based methods for monocular depth estimation, encompassing supervised, self-supervised, and semi-supervised models. Section IV introduces publicly available benchmark datasets and evaluation metrics. Section V discusses the challenges and prospects for the development of monocular depth estimation. Section VI concludes the whole review.

II. Monocular Depth Estimation Based on Traditional Methods

The traditional methods for estimating depth rely on depth cues and require specific equipment, such as stereo cameras and configuring input [29]. Meanwhile, we categorize the machine learning methods for depth estimation as traditional methods. Despite the infrequent application or utilization in recent years, these traditional methods have yielded favorable outcomes for depth estimation and continued to serve as guiding principles and valuable resources for the current mainstream research.

A. *Depth cues-based method for monocular depth estimation*

In the realm of depth cues-based methods for monocular depth estimation, shape-from-x (where x can denote stereo, motion, texture, or shading) is a well-established technique for extracting 3D information from monocular 2D images. While binocular vision is typically relied upon for depth perception, the human brain also employs various heuristic cues that can be extracted from a single perspective. Monocular depth cues mainly include focusing/defocusing [12], linear perspective [13], shadow [14], texture [30], occlusion [31], and relative height [32]. The shadow-based cue, also known as shape from shading (SFS) [33], which computes the 3D shape of a surface from one image of that surface (Fig. 2). The goal of SFS is to estimate the height function of a 3D surface of the object, given knowledge of its brightness source position and surface reflectance, such that the resulting 2D image matches a specified intensity image [34].

Additionally, this paper introduces structure from motion (SfM), which utilizes camera movement to determine the spatial and geometric relationships of targets [35]. The 3D structure and the internal and external parameters of the camera are also reconstructed through the correspondence of 2D feature points across video frames. The complete pipeline of the incremental SfM [36] is illustrated in Fig. 3. At the outset, SfM identifies suitable disordered images and extracts camera focal length information from them. Subsequently, the SIFT algorithm [16] is used to extract image features and compute Euclidean distances between feature points of two images for point matching. Image registration and triangulation techniques are employed to progressively reconstruct the 3D objects. The core of SfM lies in the Bundle Adjustment (BA) process, which involves performing an initial BA for two selected images and then incorporating new images iteratively until no further images can be integrated into the BA process. Ultimately, this leads to the estimation of camera parameters and scene geometry information, specifically resulting in a reconstruction of the sparse 3D point cloud.

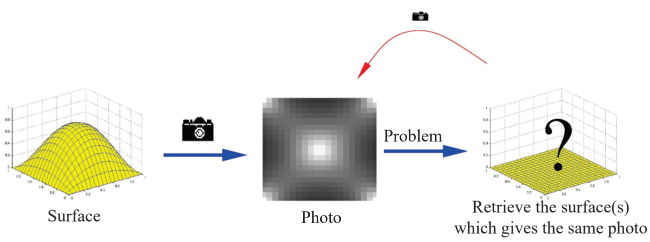


Fig. 2. The shape of shading (SFS) problem, as mentioned in [33], is a method for computing the 3D shape of a surface based on the brightness information extracted from a single black and white image of that surface.

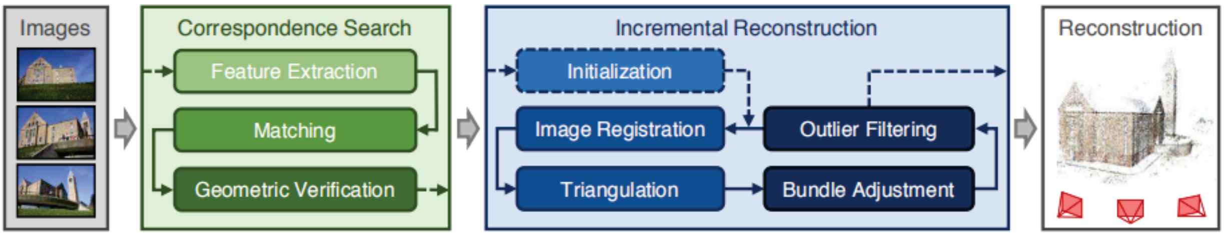


Fig. 3. The complete pipeline of the incremental structure from motion (SfM), as cited in [36], mainly consists of three subtasks: correspondence search, incremental reconstruction, and final reconstruction.

B. Machine learning method for monocular depth estimation

The depth estimation algorithms based on statistical models have been well researched due to the non-restrictive scene conditions and high applicability. This type of algorithm primarily employs machine learning methods by feeding a large number of representative training image sets and corresponding depth sets into the defined models for supervised learning. Upon completion of the training, actual images are inputted into the trained models for accurate depth prediction.

Monocular depth estimation methods based on machine learning can be divided into parametric learning methods and non-parametric learning methods. The parameter learning methods refer to techniques that involve unknown parameters in the energy function, and the training processes aim to solve these parameters. Saxena et al. [37] introduced the MRF model to learn the mapping relationship between input image features and output depth map. The above method requires an artificial assumption that the correlation between RGB images and depth maps conforms to a specific parameter model, yet this assumption is challenging to replicate the mapping relationship in real-world scenarios, thereby limiting prediction accuracy. Non-parametric learning methods utilize existing datasets for similarity retrieval to predict the depth without requiring parameter-based learning. Concurrently, non-parametric learning-based depth estimation algorithms represent a category of data-driven approaches. Given a test image, the methods can infer the depth of the test image by fusing the depth of images similar to its 3D scene content in the RGB-D databases [38]–[41].

III. Monocular Depth Estimation Based on Deep Learning Methods

Deep learning is a type of machine learning algorithm which employs multiple layers to extract high-level features from raw input data [42]. With the continuous improvement and iterative update of various deep learning models, it has made remarkable achievements in image classification, object detection and recognition, data mining, multimedia learning, natural langue processing (NLP), autonomous driving, and other domains. Similarly, in recent years, monocular depth estimation methods based on deep learning have manifested their effectiveness through the use of various neural network architectures, such as convolutional neural networks (CNNs) [43], recurrent neural networks (RNNs) [44], variational auto-encoders (VAEs) [45], and generative adversarial networks (GANs) [46]. Depth prediction based on a deep neural network was first proposed by Eigen et al. [20] in 2014. Afterward, a sequence of depth estimation methods integrated with deep learning was presented [47]–[52].

Considering the various manners of training, we categorize deep learning-based methods for monocular depth estimation into supervised, self-supervised, and semi-supervised models. Additionally, we will provide a detailed overview of several network architectures with state-of-the-art (SOTA) performance for each training manner.

A. Supervised model for monocular depth estimation

In the context of monocular depth estimation, supervised models require a single RGB image and the corresponding depth map obtained from depth cameras or laser radar sensors as ground truth (GT) for training in a supervised manner. Supervised models for monocular depth estimation differ from previous methods in that they can learn directly from raw pixel data without the need for handcrafted features. The general working flow of the supervised model for monocular depth estimation is summarized in Fig. 4.

Eigen et al. [20] introduced a multi-scale convolutional model, where a coarse-scale network first predicted the depth of scenes and then a fine-scale network refined these predictions within local regions. The architecture of the network can be seen in Fig. 5. Several studies have adopted this approach by introducing scene prior information to obtain surface normal predictions [53], improving accuracy by conditional domain randomization, or transforming the regression

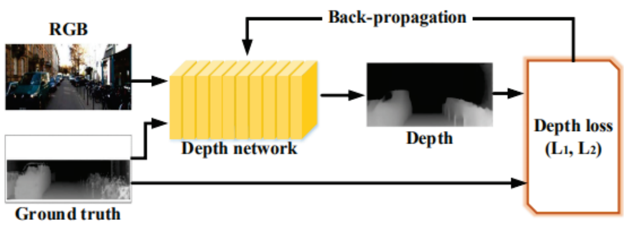
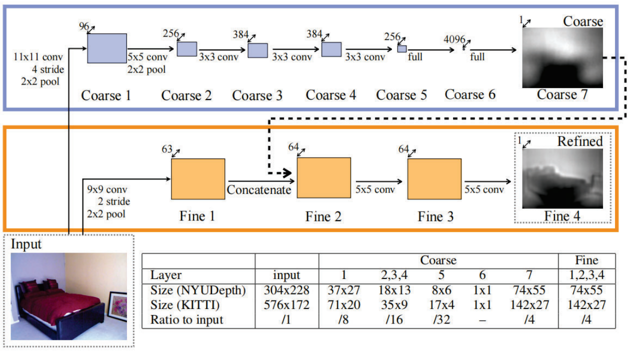


Fig. 4. The general working flow of the supervised model for monocular depth estimation, as outlined in [8], where the RGB and GT images are provided as inputs and the estimated depth map is generated as output.



		Coarse					Fine
Layer	input	1	2,3,4	5	6	7	1,2,3,4
Size (NYUDepth)	304x228	37x27	18x13	8x6	1x1	74x55	74x55
Size (KITTI)	576x172	71x20	35x9	17x4	1x1	142x27	142x27
Ratio to input	/1	/8	/16	/32	–	/4	/4

Fig. 5. Model architecture of the network proposed by Eigen et al. [20], which integrated the coarse-scale and fine-scale networks for depth estimation.

problem into a classification problem [54]. For instance, the BTS network [22] recovered the depth based on the local plane assumption, progressively refining the depth map layer by layer to achieve high accuracy. Subsequent works such as NeWCRFs [24], GLPDepth [55], and Binsformer [56], have followed the BTS approach to achieve SOTA performance.

B. Self-supervised model for monocular depth estimation

In the self-supervised model for depth estimation, the algorithms typically depend on a pair of frames as input and are trained on continuous frames or videos. This is because binocular disparity information collected by binocular cameras plays a crucial role in reasoning depth through self-supervised learning. The main advantage of the self-supervised models for monocular depth estimation lies in the capacity to facilitate training with lower-quality data, rather than solely focusing on enhancing results [29]. In addition, the self-supervised models are trained by leveraging sparsely annotated portions of the dataset and creating new data. Fig. 6 shows the architecture of the self-supervised model for depth estimation.

The typical self-supervised algorithms are MonoDepth and its variants [57]–[59]. Under several fundamental assumptions of the algorithm, including stationary objects, moving cameras, and consistent image brightness, the depth is predicted by deep networks, and the camera motion is predicted by pose

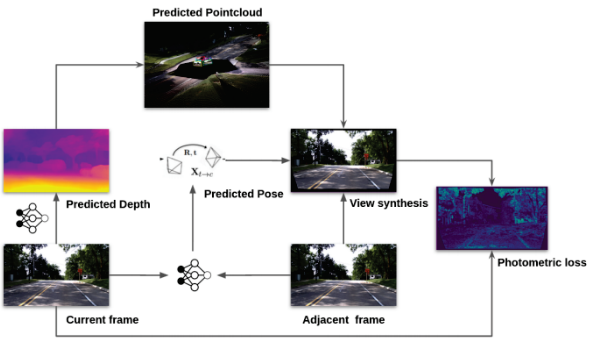


Fig. 6. The architecture of predicting depth maps based on the self-supervised model, as referenced in [29].

networks. The image of the intermediate frame is projected onto a 3D space and then re-projected back to a 2D plane for the purpose of optimizing depth prediction by calculating the re-projection error of both front and rear frames.

Furthermore, numerous SOTA self-supervised methods have been exploited. Poggi et al. [60] introduced a self-supervised-based depth estimation method by incorporating uncertainty modeling into the training paradigm. Shao et al. [61] built an enhanced self-supervised framework for monocular depth estimation, which utilized the learned appearance flow to align brightness conditions across adjacent frames. The optimized model surpassed existing methods by a large margin on the SCARED dataset [62]. Liu et al. [63] proposed a novel self-supervised monocular depth estimation method, using self-reference distillation to provide supervision signals for the student model. The experimental results on the KITTI and Make3D datastes [64], [65] showed that the method outperformed previous SOTA competitors.

C. Semi-supervised model for monocular depth estimation

To enhance the efficacy of using unlabeled datasets that are readily available in the public domain, as well as datasets that can be easily captured by small teams, semi-supervised learning models are employed to optimize learning performance [29]. The semi-supervised models combine the strengths of supervised and self-supervised models by leveraging both labeled and unlabeled data. The supervised learning part is basically the same as the supervised method, employing depth GT values for supervision. The unsupervised part is similar to the self-supervised method, and the relationship between adjacent frames is used for supervision.

In semi-supervised monocular depth estimation, Chen et al. [66] proposed a novel way to learn depth prediction in unconstrained images using relative depth information and a depth ranking loss function. Shanshan et al. [67] proposed GASDA, a semi-supervised method that leveraged geometry-aware symmetric domain adaptation to estimate depth maps. Intending to enhance the generalizability of depth estimation methods, this approach estimates depth from natural images by training the model on synthetic data. Subsequently, Kuznietsov et al. [68] introduced a semi-supervised learning method to train the network while using sparse radar depth data for direct supervision and using image alignment loss as the second training object. Different from the work of Kuznietsov et al. [68], Amiri et al. [69] achieved semi-supervised monocular depth estimation by utilizing a loss function that explicitly leveraged left-right consistency in stereo reconstruction [70], an approach not previously employed in semi-supervised training. The proposed architecture could achieve SOTA performance in depth prediction accuracy.

IV. Publicly available benchmark datasets and evaluation metrics

A. Datasets for monocular depth estimation

In the field of monocular depth estimation, several crucial datasets are highly favored due to their provision of images from diverse viewpoints and corresponding depth maps. Consumer-grade sensors such as Kinect cameras and laser radars are frequently utilized for capturing GT images in these datasets. Detailed information is presented below.

KITTI: The KITTI dataset [6] is the largest and most widely used dataset for various sub-tasks in computer vision, such as object detection, depth estimation, and semantic segmentation. The dataset is accessible in two versions and comprises 394 road scenes, providing RGB stereo scenes and corresponding GT depth maps. The KITTI dataset contains images from urban, residential, and road scenes, where the 56 scenes are divided into two parts, half for training and the other half for testing, by Eigen et al. [20]. Each scene is composed of stereo image pairs with a resolution of 1224×368. The GT depth images are captured by the Velodyne laser radar.

NYU Depth: The NYU Depth dataset [71] focuses on indoor environments with 464 scenes. Unlike the KITTI dataset that employs laser radar for GT value collection, the NYU Depth dataset uses Kinect cameras to capture monocular video sequences and scene depth values for GT generation. The training and testing datasets consist of 249 and 215 scenes respectively, with RGB sequence images having a resolution of 640×480 pixels; half of the samples are downsampled during the experimentation.

Make3D: The Make3D dataset [65] exclusively consists of monocular RGB images and depth images without stereo counterparts. It encompasses 400 training and 134 testing outdoor images, as well as diverse outdoor, indoor, and synthetic scenes used for depth estimation.

SceneFlow: The SceneFlow dataset [72] is presented as the first large-scale synthetic dataset, consisting of 39K stereo images with corresponding disparity, depth, optical flow, and segmentation masks.

Cityscapes: The Cityscapes dataset [73] comprises 5,000 images with fine annotations and 20,000 images with coarse annotations. Meanwhile, it consists of a series of stereo videos collected from 50 cities over the course of several months. The training dataset comprises 22,973 stereo image pairs with a resolution of 1024×2048. As this dataset lacks GT depth information, it is only suitable for the training of various unsupervised depth estimation methods.

B. Evaluation Metrics

The depth estimation methods are evaluated and compared using specific evaluation metrics proposed by Eigen et al. [20]. The evaluation indicators include root mean square error (RMSE), the logarithm root mean square error (RMSE log), absolute relative error (Abs Rel), square relative error (Sq Rel), and the accuracy rate metrics include $\delta < 1.25^t$, where t = 1,2,3. These metrics are formulated as follows:

RMSE = $\sqrt{\frac{1}{|N|}\sum_{i\in N}\|d_i - d_i^*\|^2}$,

RMSE log = $\sqrt{\frac{1}{|N|}\sum_{i\in N}\|\log(d_i) - \log(d_i^*)\|^2}$,

Abs Rel = $\frac{1}{|N|}\sum_{i\in N}\frac{|d_i - d_i^*|}{d_i^*}$,

Sq Rel = $\frac{1}{|N|}\sum_{i\in N}\frac{\|d_i - d_i^*\|^2}{d_i^*}$,

Accuracies: $\%$ of d_i s.t. $\max\left(\frac{d_i}{d_i^*}, \frac{d_i^*}{d_i}\right) = \delta < thr$,

where d_i is the prediction depth value of pixel, and d_i^* is the GT of depth. N represents the total number of pixels with real-depth values, and thr is the threshold.

V. Current Challenges and Prospects

In recent decades, monocular depth estimation has achieved significant breakthroughs and advancements, particularly in conjunction with deep learning methods. Nevertheless, there remain several challenges that must be confronted.

(1) The fields of autonomous driving, field robotics, and large-scale scene reconstruction necessitate the use of very-long range datasets. However, current research primarily focuses on indoor and short-distance outdoor datasets as well as those collected from moving vehicles. This lack of remote data also poses challenges for extrapolation or adaptation.

(2) In multi-task learning, deep learning methods for monocular depth estimation typically employ multiple sub-networks or sub-modules to handle distinct sub-tasks, resulting in a certain degree of increased computational resource and memory consumption.

(3) Complex deep neural networks exhibit scalability in terms of memory requirements, which is a significant challenge when processing high-resolution images and predicting high-resolution depth images.

(4) Influenced by complex scenes such as occlusion, highly cluttered environment, and complex material properties of objects, another challenge is how to achieve greater accuracy.

Consequently, multiple promising prospects are presented for the development of monocular depth estimation.

(1) Monocular depth estimation using physics-driven deep learning methods.

Recently, the physics-driven deep learning methods have demonstrated significant potential on high dynamic range imaging [74], [75], single image dehazing [76], and other areas [77], particularly in reducing the reliance on extensive precomputed training data. In the future, physics-driven deep learning methods can be employed to reduce uncertainties for monocular depth estimation.

(2) Simultaneous high dynamic range imaging and monocular depth estimation.

Existing monocular depth estimation algorithms assume all regions of input images are well exposed, which is not applicable to scenes with high dynamic range [78]. Saturated regions pose challenges to the existing monocular depth estimation algorithms. By combining with high dynamic range imaging, a greater dynamic range can be obtained than that captured by a single image, resulting in a significant reduction of saturation. The monocular depth estimation can be further enhanced with more valuable visual information and improved conditions.

(3) Monocular depth estimation for adverse weathering conditions.

Generally, images captured under adverse weathering conditions such as hazy images [79] and rainy images [80], exhibit poor and varied visual quality, making it hard for monocular depth estimation. Further research can be conducted on monocular depth estimation in these scenes, which would bring an impact on enhancing the universality and robustness of depth estimation for outdoor vision systems. For instance, the resultant algorithm could be well applied to develop a vision-based flood detection system.

VI. CONCLUSIONS

Monocular depth estimation plays a crucial role in scene understanding and 3D reconstruction, with applications extending to various fields of computer vision. In this review, we present an overview of models and algorithms for monocular depth estimation, including both traditional methods and deep learning-based approaches. We provide detailed descriptions of supervised, self-supervised, and semi-supervised models to demonstrate the performance of monocular depth estimation using these SOTA methods. Additionally, publicly available benchmark datasets and evaluation metrics are presented. Finally, we summarize the current challenges and promising prospects for the development of monocular depth estimation.

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